

# Tools in production

LESSON 02 · EQUIP & CONNECT – STUDIO: TOOLS, MODELS & MCP

## What you'll learn

- Securely authenticate custom tools using static API keys or dynamic OAuth 2.0 flows.
- Configure the secure ACI (Agent Connector Integration) card built on open-source `aci.dev` to handle refresh tokens automatically.
- Limit agent overhead by trashing unneeded actions from prebuilt integrations (like Slack's 800 actions).
- Balance open-ended agent autonomy against structured paths to optimize speed, context size, and token costs.

## Key concepts

**Most production APIs require secure authentication like static keys or OAuth.** Unlike public testing endpoints, production tools require credentials. Studio allows you to seed API keys directly in the default query parameters, keeping them completely hidden from both the LLM and the end-user.

**OAuth 2.0 is fully managed by Lyzr's integration infrastructure.** For complex user authorization (e.g., Gmail, Salesforce), Lyzr stores client IDs/secrets and automatically handles token refreshes, preventing connection timeouts when tokens expire.

**Trimming unneeded actions is the easiest way to cut agent latency and token cost.** A standard integration like Slack carries hundreds of default actions. Activating all of them inflates the prompt context window and causes reasoning lag. Deselecting unused actions optimizes performance.

**Autonomy is a dial you tune based on the task.** If the agent is doing open-ended task solving, let it carry a wide range of actions. If the agent runs on a fixed, known road, trim it down to only the exact actions required to do the job.

## In Studio

Studio feature: **Build → Custom Tool** and **Connections → Integrations**. Here's how to configure authenticated tools and limit active actions:

1. Go to **Build** and select **Custom Tool**. Choose **OpenAPI** and paste your spec.
2. In the configuration fields, locate **Default Query Parameters** (or headers) and insert your API key once. The key will carry automatically with every request.
3. For OAuth setups, choose the **Custom ACI Tool** card. Input your Client ID, Client Secret, Scope list, Authorize URL, Token URL, and Refresh URL.

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4. Attach a prebuilt integration (e.g., Slack) to your agent. Open the integration's settings panel.
5. Review the active list of actions. Uncheck the checkboxes next to actions the agent does not require to do its job.
6. Click **Save** to deploy the authenticated, lightweight tool suite.

## TRY THIS

- 1** Create a custom tool that uses a public API requiring an API key (e.g., NASA Picture of the Day API using `DEMO_KEY`). Verify that the key is stored in the default query parameters and does not leak in the chat traces.
- 2** Attach the **Slack** integration to an agent. Go to its action registry and deselect everything except `post_message` and `read_message`. Observe the change in available tool definitions.
- 3** Contrast a "full-roam" agent with a "trimmed" agent—explain when you would use each.